

Problem 49 (IberoAmerican MO 2019 Problem 2). Find all nonzero monic polynomials $P(x)$ with integer coefficients such that its roots are exactly $P(0), P(1), \dots, P(n-1)$ (counted with multiplicity) for some positive integer n .

Solution. The only solution is $P(x) = x$, which clearly satisfies the condition.

The condition means

$$P(x) = (x - P(0))(x - P(1)) \cdots (x - P(n-1)). \quad (1)$$

Putting $x = 0$ in (1), we obtain $P(0) = (-1)^n P(0)P(1) \cdots P(n-1)$. This gives $P(0) = 0$ or $P(1) \cdots P(n-1) = (-1)^n$.

If $P(0) = 0$, we may assume $n \geq 2$ (since $n = 1$ implies $P(x) = x$). Putting $x = 1$ in (1), we get

$$P(1) = (1 - P(1)) \cdots (1 - P(n-1)). \quad (2)$$

If $P(1)$ is odd, then the left-hand side of (2) is odd but the right-hand side is even (as $1 - P(1)$ is even), which is impossible. If $P(1)$ is even, as $P(0) = 0$ is also even, this shows $P(x)$ is even for any integer x . However, the left-hand side of (2) is even but the right-hand side is odd, which is impossible.

If $P(0) \neq 0$ and $P(1) \cdots P(n-1) = (-1)^n$ (which implicitly shows that $n \geq 2$), we must have $P(k) = \pm 1$ for each $k = 1, 2, \dots, n-1$. If $P(k) = 1$ for some $k = 1, 2, \dots, n-1$, then by putting $x = 1$ in (1), we get $P(1) = 0$ since $1 - P(k) = 0$. Putting $x = 0$ in (1), we get $P(0) = 0$, contradiction. Therefore, we must have $P(k) = -1$ for all $k = 1, 2, \dots, n-1$. However, putting $x = 1$ in (1) gives $-1 = (1 - P(0))2^{n-1}$, which cannot be true as $n \geq 2$.

Therefore, the only solution is $P(x) = x$. □

Problem 50 (USAJMO 2016 Problem 2). Prove that there exists a positive integer n not exceeding 1000000 satisfying the following property: the decimal representation of 5^n has 6 consecutive zeros.

Solution. We claim that $n = 2^{19} + 20 = 524308$ satisfies all conditions.

Firstly, we have $n < 1000000$. Secondly, by the Euler-Fermat theorem, we have $5^n \equiv 5^{20} \pmod{2^{20}}$ since $\varphi(2^{20}) = 2^{19}$. Also, it is clear that $5^n \equiv 0 \equiv 5^{20} \pmod{5^{20}}$. Thus, we have $5^n \equiv 5^{20} \pmod{10^{20}}$. It follows that the last 20 digits of 5^n are the same as those of 5^{20} .

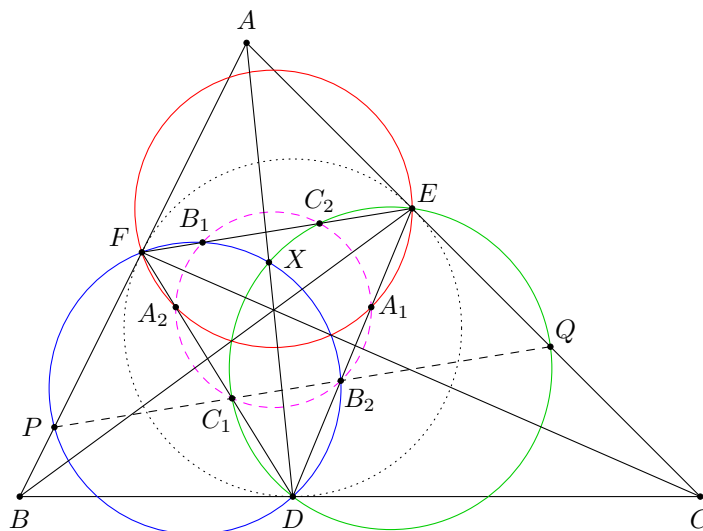
Now, we see that $5^{20} = (125)^2 \cdot 5^{14} < (128)^2 \cdot 5^{14} = 2^{14} \cdot 5^{14} = 10^{14}$. This shows 5^{20} is at most a 14-digit number, and so the last 20 digits of 5^{20} start with 6 consecutive zeros. Combining all these, we know that 5^n contains 6 consecutive zeros. □

Problem 51 (IberoAmerican MO 2011 Problem 3). In $\triangle ABC$, let D, E, F be the contact points of the incircle with the sides BC, CA, AB respectively. Let Γ_1 be a circle passing through E and F , Γ_2 be a circle passing through F and D , Γ_3 be a circle passing through D and E , such that Γ_2 and Γ_3 meet again on the line AD , Γ_3 and Γ_1 meet again on the line BE , Γ_1 and Γ_2 meet again on the line CF . Suppose Γ_1 intersects the segments DE and DF at A_1 and A_2 respectively, Γ_2 intersects the segments EF and ED at B_1 and B_2 respectively, Γ_3 intersects the segments FD and FE at C_1 and C_2 respectively. Prove that $A_1, A_2, B_1, B_2, C_1, C_2$ are concyclic.

Solution. Let X be the intersection point of Γ_2 and Γ_3 different from D . Let P be the second intersection point of Γ_2 and the line AB . Let Q be the second intersection point of Γ_3 and the line AC .

Firstly, since $AF \times AP = AX \times AD = AE \times AQ$ and $AF = AE$, we have $AP = AQ$. Thus, using directed angles, we obtain

$$\angle QPF = \angle QPA = \frac{\pi}{2} - \frac{A}{2} = \angle EDF = \angle B_2DF = \angle B_2PF.$$



This implies B_2 lies on PQ . Similarly, C_1 lies on PQ . It follows that

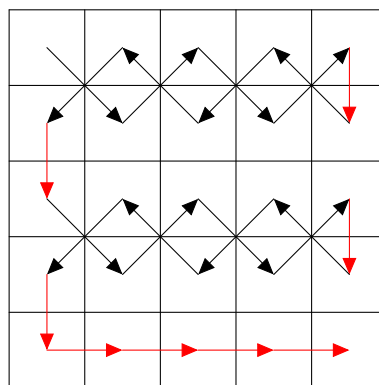
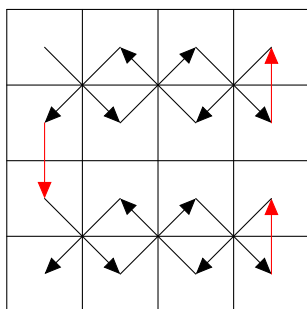
$$\angle C_1 B_2 D = \angle P B_2 D = \angle P F D = \angle B F D = \angle F D B = \angle F E D = \angle F E A_1 = \angle F A_2 A_1 = \angle C_1 A_2 A_1.$$

This shows A_1, A_2, C_1, B_2 are concyclic. By symmetry, B_1, B_2, A_1, C_2 are concyclic, and C_1, C_2, B_1, A_2 are concyclic. If the three circles containing these points are pairwise distinct, then their pairwise radical axes $A_1 B_2, B_1 C_2, C_1 A_2$ are concurrent or parallel, which is clearly not the case. Therefore, two of these circles are the same, which implies $A_1, A_2, B_1, B_2, C_1, C_2$ are concyclic. $\square$

Problem 52 (IberoAmerican MO 2019 Problem 5). The king is placed in one of the cells of an $n \times n$ chessboard, where n is a positive integer. In each move, the king can move to an adjacent cell sharing at least one common vertex. Consider all possible paths of the king such that each cell is visited exactly once (the initial cell should not be visited again). What is the largest possible number of moves that the king moves diagonally?

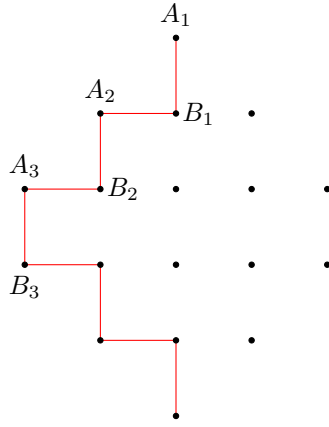
Solution. The answer is $n(n-1)$ if n is even, and is $(n-1)^2$ if n is odd.

The constructions for $n = 4$ and $n = 5$ are given below, which can be easily generalized. For even n , only the $n-1$ vertical moves are non-diagonal, and so there are $n^2 - 1 - (n-1) = n(n-1)$ diagonal moves. For odd n , only the $n-1$ vertical moves and the $n-1$ horizontal moves are non-diagonal, and so there are $n^2 - 1 - 2(n-1) = (n-1)^2$ diagonal moves.



We now prove the upper bound. We first consider the case when n is even. Using chessboard colouring, note that a diagonal move must move to a cell of the same colour, while a non-diagonal move must move to a cell of distinct colour. Thus, we can partition the whole path into monochromatic paths such that the number of non-diagonal moves is the number of monochromatic paths minus 1. It suffices to minimize the number of monochromatic paths.

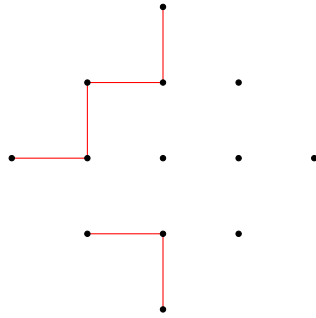
Consider only the black cells. We use the centre of the black cells to represent them. For example, the figure below shows all points representing the black cells when $n = 6$ (the chessboard is rotated by 45°). A diagonal move in the original chessboard corresponds to an upward, downward, leftward or rightward move in this figure. We label the topmost points in each column in the left half as $A_1, A_2, \dots, A_m$ as shown where $m = \frac{n}{2}$. Also, we label the point below A_j as B_j .



Recall that we have to partition these black points into some paths, and we need to minimize the number of paths. We start with any partition that minimizes the number of paths (which exists since there are finitely many partitions). If A_1 is not adjacent to B_1 , then A_1 is the only point in its path. We may connect A_1 and B_1 , and possibly remove an edge incident on B_1 . This does not increase the number of paths. Similarly, if A_2 and B_1 are not adjacent, then A_2 must be an endpoint of its path. So connecting A_2 and B_1 (while keeping the edge between B_1 and A_1) does not increase the number of paths. In the same way, we may assume A_2 and B_2 are adjacent, A_3 and B_2 are adjacent, and so on.

Eventually, we may assume some pairs of points are adjacent, as shown by the red edges in the figure. Note that this red path cannot be further extended. Removing this red path, we are reduced to the case $n - 2$. Thus, by induction, it is easy to see that there are at least m paths among the black points. By symmetry, there are at least m paths among the white points. From above, there are at least $m + m - 1 = n - 1$ non-diagonal moves, and so at most $n^2 - 1 - (n - 1) = n(n - 1)$ diagonal moves.

The same argument applies to the odd case. The figure below shows the case of $n = 5$, where the corner cells of the original chessboard are black. Following the above arguments, we can only assume the outermost points belong to two red paths as shown. Removing these red paths, we are reduced to the case $n - 2$. Thus, by induction (with base case $n = 1$ having 1 path), there are at least n paths among the black points.



In order to go from one black path to another, one must go through a white path. Therefore, there are at least $n - 1$ white paths. In total, there are at least $2n - 1$ paths, which means at least $2n - 2$ non-diagonal moves. Thus, the number of diagonal moves is at most $n^2 - 1 - (2n - 2) = (n - 1)^2$. This completes the proof. $\square$